Benchmarking FSRS against a Continuous-Time Bayesian Retention Model on Anki Review Logs

Evaluation protocol, implementation and results

Final Year Project — Axiom (`BayesianScheduler` branch)

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Abstract

We build a reproducible, leakage-free evaluation pipeline that compares the Free Spaced Repetition Scheduler (FSRS-6) with our continuous-time Bayesian dynamic retention model (DLM) and with Half-Life Regression (HLR, Settles & Meeder 2016). All models are scored on exactly the same 95,400 held-out review events from 100 users of the public `anki-revlogs-10k` dataset, using a per-user time split and strictly one-step-ahead prediction. Each card's memory state is reconstructed by replaying its review history in chronological order. On log loss, Brier score, AUC and calibration (ECE), FSRS-6 with per-user optimised parameters is the best model overall (log loss 0.379, AUC 0.777, ECE 0.009) and in every review-gap bin. Our model (log loss 0.422, AUC 0.692, ECE 0.025) is significantly worse than both FSRS-6 variants under a user-clustered paired bootstrap, but far better than HLR (log loss 0.680, AUC 0.555). Adding the practice-dependent decay term κ changes nothing measurable. The Gibbs sampler did not converge at 2,000 iterations ($\hat{R}_\lambda \approx 2.4$), so the results for our model are provisional. We give evidence that its predictions are nonetheless insensitive to the poorly mixed parameters.

1 Introduction

Spaced repetition systems schedule reviews from a predicted probability of recall. Our project proposes a continuous-time Bayesian dynamic linear model (DLM) of human retention, in which a latent memory state decays between reviews and is updated by each outcome. Earlier project updates compared the DLM with static baselines: a lagged logistic regression and HLR. This report adds the comparison that a reviewer will expect first: FSRS, the open-source scheduler used by Anki and the strongest widely used public baseline.

The task was to:

1. implement a working evaluation pipeline for FSRS on the `anki-revlogs-10k` dataset (Open Spaced Repetition);
2. reconstruct each card's memory state by replaying its review history chronologically with the official FSRS implementation;
3. predict recall probabilities for held-out reviews without data leakage;
4. compare FSRS with our model on exactly the same test events, using log loss, Brier score, AUC and calibration metrics;
5. document the protocol, implementation choices and results, with reproducible code.

HLR, the model of the Duolingo paper, is included as a third reference point.

2 Data

Source. `anki-revlogs-10k` is a gated Hugging Face dataset with one review log per Anki user. Each row is one review, with the card id, the day offset, the rating (1 = Again, 2 = Hard, 3 = Good, 4 = Easy), the card state and the elapsed time. We sample 100 users with a fixed seed (0), and at most 1,000 cards per user, sampled with the same seed. The cap keeps the Gibbs sampler tractable. A 5-user pilot showed that 1,000 cards per user, unlike 300, leaves almost every user with enough data for the FSRS optimiser.

Event definition.

- Rows with rating 0 (manual reschedules) are dropped.
- Filtered-deck reviews (state 4) are *kept*. They carry real ratings and durations, and dropping them gave FSRS wrong gaps (in one case 515 days instead of 1).
- Cards whose first kept review is not a new-card review are dropped, because their history is truncated and their memory state cannot be rebuilt.
- The label is $y = 1$ (recalled) if the rating is above 1.
- Review gaps are recomputed over the kept rows, so every model sees identical gaps.

Table 1: The full run (`results/full_u100_c1000`).

Users	100	Train events	606,364
Cards	99,074	Test events	162,294
Review events	768,658	Scored test events	95,400
		Recall rate (scored)	0.831

3 Evaluation protocol

Split. For each user, the cutoff is the 80th percentile of that user's review days. Reviews before the cutoff are TRAIN, and the rest are TEST. All model parameters are fitted on TRAIN rows only and frozen during the test period.

Scored events. A TEST review is scored if it is not the card's first review and its gap is at least one day. This is the srs-benchmark convention of scoring inter-day reviews. Same-day and first reviews are still fed to the models as history, but are never scored.

One-step-ahead prediction. To predict scored event k, a model may use the card's entire history before k, including earlier test reviews and their outcomes. It may not use event k or anything after it. This mirrors how a scheduler is used in practice.

Leakage tests. The pipeline has 260 automated tests. Those that enforce the protocol check that:

- no TRAIN event lies at or after a user's cutoff;
- changing a scored event's own outcome, or any later outcome, never changes its prediction (checked for every model);
- flipping every TEST label leaves all five models' predictions identical;
- users and cards seen only in TEST use our model's cold-start path;
- all five models predict exactly the same set of event ids.

4 Models

FSRS-6 (default and optimised). We use the official `py-fsrs` 6.3.2. Each card's reviews are replayed in order through the `Scheduler`, with fuzzing disabled. Before each scored review, the card's retrievability at that moment is read with `get_card_retrievability`, and this is the prediction. Only then is the review applied. The *default* variant uses the published default weights. The *optimised* variant fits the `py-fsrs` `Optimizer` per user on that user's TRAIN reviews. `py-fsrs` silently keeps the default weights when a user has fewer than 512 informative reviews. We apply that rule explicitly: 1 of the 100 users (292 informative train reviews) falls back to the defaults. The other users have a median of 2,674.

HLR (Settles & Meeder 2016). The recall probability is $p = 2^{-\Delta t/h}$, with half-life $h = 2^{\theta^\top x}$ and features $x = (\sqrt{1+n_{\text{right}}},\ \sqrt{1+n_{\text{wrong}}},\ 1)$ counted from the card's prior reviews. Training is a faithful port of the SGD update in Duolingo's official `experiment.py`, using the paper's loss, half-life and probability clipping, and per-feature learning rates. It was checked against the official arithmetic to 10^{-12}. Anki has no lexeme features, so they are omitted.

Ours: continuous-time Bayesian DLM. The unmodified `DynamicRetentionModelDT` is used through an adapter. The latent memory state $\psi_{n,t}$ of card n reverts towards a hierarchical mean with a gap-dependent persistence:

$$\phi_{n,t} = \exp(-\lambda_{n,t}\,\Delta t_{n,t}), \qquad \lambda_{n,t} = \lambda \exp(-\kappa\, C_{n,t}), \qquad \mu_{n,t} = b_0 + \alpha_u + \beta_i + \gamma_u X_{n,t},$$

where $C_{n,t} = \log(1 + \text{successes})$ and $X_{n,t}$ is the lagged outcome. Outcomes are Bernoulli with a logit link. The posterior is sampled with Pólya-Gamma augmentation and forward-filtering backward-sampling (FFBS) Gibbs sweeps: 2,000 iterations, 1,000 burn-in, thinning 4, seed 0. Predictions are the model's own one-step filtered recall probabilities, averaged over posterior draws. Two variants are run: learned λ with $\kappa = 0$, and learned λ with learned κ.

5 Metrics

With N scored events, labels y_j and predictions p_j:

- **Log loss:** $-\frac{1}{N}\sum_j [y_j \log p_j + (1-y_j)\log(1-p_j)]$, with p clipped to $[10^{-6}, 1-10^{-6}]$.
- **Brier score:** $\frac{1}{N}\sum_j (p_j - y_j)^2$.
- **AUC:** rank-based (Mann–Whitney), with ties counted as one half.
- **ECE:** 10 equal-width bins, $\sum_b \frac{n_b}{N} |\bar{p}_b - \bar{y}_b|$.
- **RMSE (bins):** the same over 20 bins.
- **Mean bias:** $\bar{p} - \bar{y}$, so a positive value means the model is over-confident.

Results are also split by review gap: 1 d, 2–3 d, 4–14 d, 15–60 d and 61 d+.

Uncertainty. Reviews from the same person are correlated, so confidence intervals use a *user-clustered paired bootstrap*. In each of 1,000 resamples, users are drawn with replacement, all of each drawn user's events are taken, both models are scored on the same resample, and the metric difference is recorded. We report 95% percentile intervals.

6 Results

6.1 Overall

Table 2: Overall results on the same 95,400 scored events. Lower is better except for AUC, and mean bias is best near zero. The best value in each column is in bold.

Model	Log loss	Brier	AUC	ECE	RMSE(bins)	Mean bias
FSRS-6 (optimised)	**0.379**	**0.117**	**0.777**	**0.009**	**0.014**	**−0.002**
FSRS-6 (default)	0.396	0.121	0.745	0.012	0.017	0.011
Ours (learned λ)	0.422	0.131	0.692	0.025	0.032	0.011
Ours ($+\kappa$)	0.422	0.131	0.693	0.026	0.032	0.011
HLR	0.680	0.170	0.555	0.146	0.176	0.060

FSRS-6 (optimised) is best on every metric (Table 2). Our model is clearly better than HLR on every metric, but worse than both FSRS-6 variants. HLR's weak discrimination (AUC 0.555) is close to the AUC of about 0.54 reported for HLR in the original paper.

6.2 By review gap

Table 3: Log loss (left) and AUC (right) by days since the card's previous review.

		Log loss ↓					AUC ↑				
Gap	n	F-opt	F-def	Ours	$+\kappa$	HLR	F-opt	F-def	Ours	$+\kappa$	HLR
1 d	16,983	**0.396**	0.409	0.423	0.423	0.942	**0.769**	0.748	0.708	0.708	0.658
2–3 d	16,080	**0.389**	0.402	0.426	0.426	0.734	**0.755**	0.730	0.676	0.677	0.637
4–14 d	29,204	**0.361**	0.369	0.398	0.398	0.555	**0.760**	0.737	0.680	0.681	0.616
15–60 d	15,626	**0.374**	0.401	0.419	0.419	0.473	**0.781**	0.725	0.703	0.705	0.599
61 d+	17,507	**0.389**	0.418	0.460	0.460	0.771	**0.817**	0.771	0.763	0.764	0.628

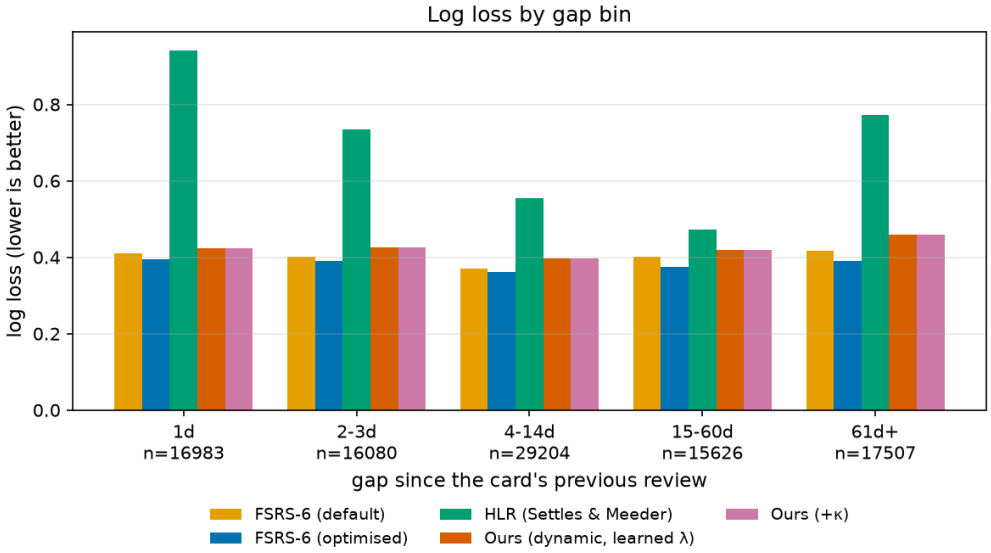


Figure 1: Log loss by gap bin (lower is better).

The ranking is the same in every gap bin (Table 3, Figure 1): FSRS-6 (optimised), then FSRS-6 (default), then our model, then HLR. Our model's log-loss gap to FSRS-6 (optimised) is widest at

long gaps (61 d+: 0.460 against 0.389). Its AUC gap is narrowest there (0.763 against 0.817).

6.3 Calibration

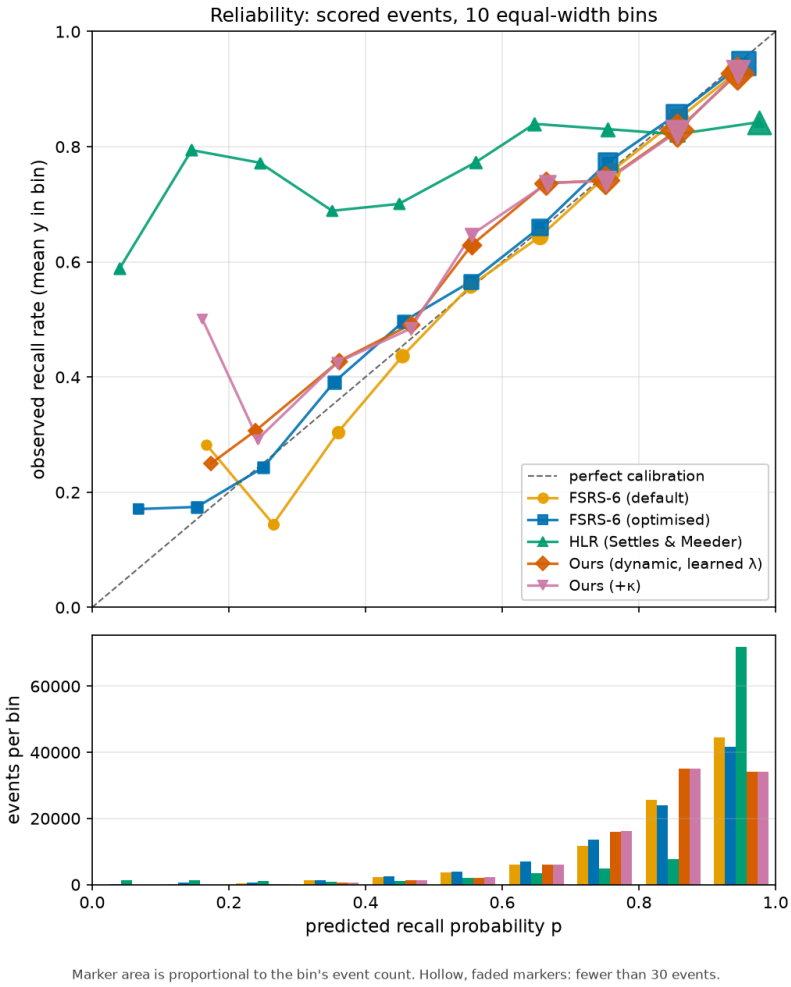


Figure 2: Reliability diagram over all gaps (10 equal-width bins). Marker area is proportional to the bin count.

Both FSRS-6 variants lie close to the diagonal (Figure 2). Our model is under-confident in the middle range: in the 0.5–0.6 bin it predicts 0.555 on average, while the observed recall is 0.629. At long gaps it becomes over-confident, with a mean bias of +0.090 at 61 d+ (ECE 0.090, against 0.027 for FSRS-6 optimised). At 1 d its bias is only −0.006. HLR is badly miscalibrated in both directions. Its mean bias is +0.170 at 1 d and −0.249 at 61 d+.

6.4 Paired differences and per-user results

Table 4: Paired differences (model minus reference) with 95% user-clustered bootstrap intervals. * marks an interval that excludes zero.

Comparison	Δ Log loss	Δ Brier	Δ AUC
FSRS-6 (def) − FSRS-6 (opt)	+0.017 [0.008, 0.025]*	+0.004 [0.002, 0.006]*	−0.032 [−0.047, −0.016]*
Ours − FSRS-6 (opt)	+0.043 [0.033, 0.055]*	+0.014 [0.011, 0.019]*	−0.084 [−0.110, −0.064]*
Ours − FSRS-6 (def)	+0.026 [0.013, 0.042]*	+0.011 [0.007, 0.015]*	−0.053 [−0.087, −0.023]*
HLR − Ours	+0.258 [0.204, 0.314]*	+0.038 [0.026, 0.052]*	−0.137 [−0.203, −0.075]*
Ours (+κ) − Ours	−0.000 [−0.001, 0.000]	−0.000 [−0.000, 0.000]	+0.001 [−0.000, 0.002]

All differences between model families are significant (Table 4). The κ term makes no detectable difference. Per user, FSRS-6 (optimised) has a lower log loss than our model for 78 of the 100 users, and FSRS-6 (default) for 68. Our model beats HLR for 86 users.

6.5 MCMC convergence

Table 5: Posterior summaries of the poorly mixing parameters (one chain, 250 kept draws). Targets are $\hat{R} < 1.01$ and an effective sample size (ESS) in the hundreds.

Variant	Parameter	Mean	95% interval	ESS	Split $\hat{R}$
Ours	λ	1.871	[1.312, 2.646]	3.0	2.44
Ours	σ^2	0.204	[0.189, 0.231]	3.3	2.07
Ours	b_0	2.332	[2.314, 2.343]	3.9	1.85
Ours (+κ)	λ	1.455	[0.898, 2.228]	3.0	2.38
Ours (+κ)	κ	−0.160	[−0.211, −0.076]	3.5	1.94

The chains did not converge at 2,000 iterations (Table 5). The decay rate λ in particular is still drifting. An attempted 8,000-iteration run saw λ move from 1.52 to 1.70 between sweeps 1,200 and 1,400. That run was cut short by a machine shutdown. The other hyper-parameters mix acceptably (τ_α^2, τ_γ^2 and g_0 have ESS of 81–250 and $\hat{R} \leq 1.06$). The λ Metropolis–Hastings acceptance rate is 0.24.

The predictions nevertheless appear insensitive to this. The two variants settle on clearly different λ posteriors (means 1.87 and 1.46), yet their predictions correlate at $r = 0.992$ over the 95,400 events, and every metric agrees to about three decimals. This suggests that the poorly mixed directions of the posterior barely move the predicted recall. It is evidence, not proof: the results for our model should be treated as provisional until a longer multi-chain run is available.

7 Duolingo dataset (full)

The same protocol was run on Duolingo's public learning traces (Settles & Meeder 2016), using all 115,222 users and all 5,861,629 (user, lexeme) cards. Each row of the file is one practice session of one lexeme, and it is one review event. The label is $y = 1$ if the recorded $p_{\text{recall}} \geq 0.5$. The split is the same per-user 80th-percentile day cutoff as on Anki. Of the 12,854,226 events, 1,189,355 are scored, with a recall rate of 0.920. The data span is about 12 days, so the 15–60 d and 61 d+ bins are empty. The per-gap tables are in `results/duolingo_full/report.md`.

Table 6: Duolingo (full dataset): overall results on the same 1,189,355 scored events. Lower is better except for AUC. The best value in each column is in bold.

Model	Log loss	Brier	AUC	ECE	RMSE(bins)	Mean bias
FSRS-6 (optimised)	0.300	0.078	0.570	0.050	0.070	−0.026
FSRS-6 (default)	0.302	0.078	0.566	0.051	0.070	−0.025
Ours (GPU port)	**0.268**	**0.071**	**0.656**	**0.014**	**0.017**	0.004
LagLogit + dt	0.291	0.076	0.653	0.020	0.068	**−0.001**
HLR	0.418	0.079	0.521	0.072	0.072	0.072

On this dataset the Anki ordering is reversed: Ours (GPU port) has the lowest log loss, and it is the only model with a clearly better AUC than FSRS-6. Against FSRS-6 (optimised), its log loss difference is −0.032 [−0.033, −0.031] and its AUC difference is +0.087 [+0.083, +0.091], both from the user-clustered paired bootstrap. LagLogit + dt differs by −0.009 [−0.012, −0.007] in log loss, and HLR by +0.118 [+0.112, +0.124]. Ours wins on log loss for 74% of the 41,223 users with finite log loss for both models.

The result is weakened by the following, and it should not be read as general evidence that our model beats FSRS.

- **FSRS-6 (optimised) is nearly the default model.** Its optimiser needs 512 informative training reviews per user, and 115,110 of 115,222 users fall back to the default weights. The difference from FSRS-6 (default) is +0.0015 in log loss.
- **64,254 users have no training rows.** Their cards are predicted from population-level parameters.
- **The label thresholds a recorded value.** y is $p_{\text{recall}} \geq 0.5$, not a single observed recall answer. FSRS sees only two grades (Good for $y = 1$, Again otherwise), so it does not use the Hard and Easy distinction.
- **HLR has no lexeme features,** so it is weaker than the published model.
- **Lambda is not identified.** At 2,000 sweeps its split $\hat{R}$ is 2.48 and its ESS is 3.0. Ours (GPU port) is the same equations as the Anki model, ported to PyTorch and run on a GPU in float64 (2,000 sweeps, 1,000 burn-in, thin 10). A run of 8,000 sweeps drifts further (lambda 1.73 at sweep 8,000, $\hat{R} = 2.33$), and σ^2 drifts with it. So lambda and σ^2 are weakly identified and should not be interpreted. The predictions barely change: test log loss is 0.2682 at 2,000 sweeps and 0.2679 at 8,000, and the two runs' predictions correlate at 0.992. The headline uses the 2,000-sweep run, as on Anki.

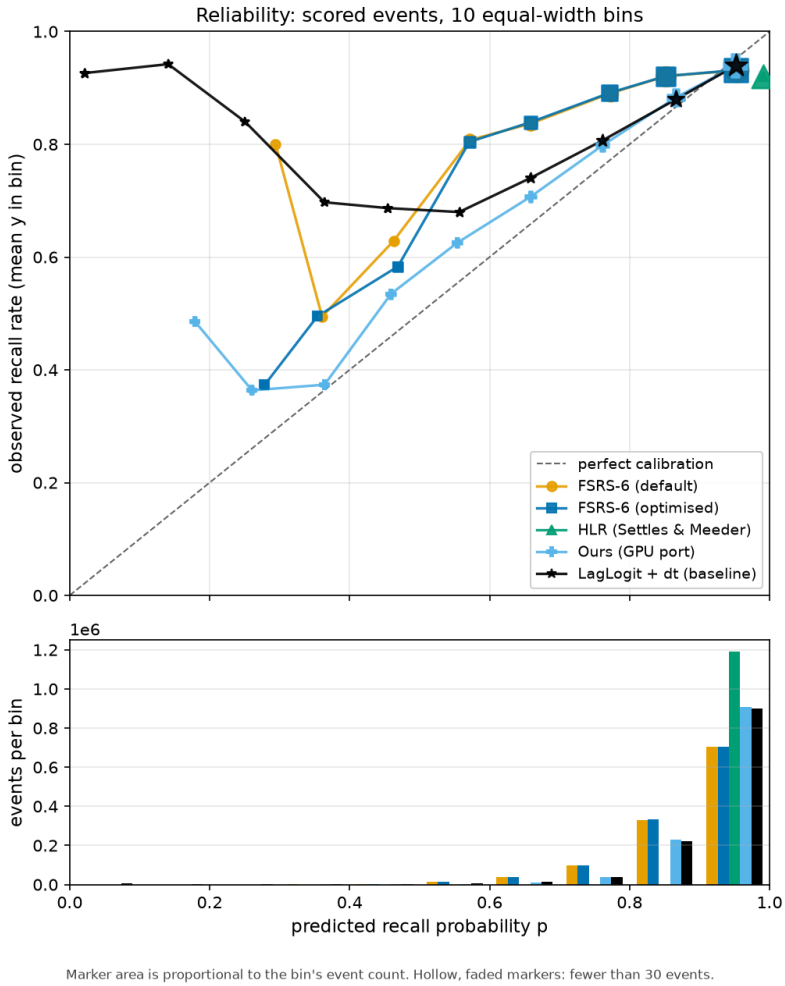


Figure 3: Duolingo (full dataset): reliability diagram over all gaps (10 equal-width bins). Marker area is proportional to the bin count.

8 Improved models (Phases 6–7)

Two further phases built models that were tested against FSRS-6 (optimised). Phase 6 added a hierarchical FSRS-style model (v2), which ties FSRS-6 on Anki and beats it on Duolingo. Phase 7 added v3 (v2 with an L2 penalty on the user-effect scales, a lagged outcome and a fixed-scale per-card effect), which ties FSRS-6 on Anki and beats it on Duolingo, and a gradient-boosted tree model (GBDT, LightGBM) on past-only features, which is the best model on both datasets. Full details are in `docs/EVALUATION.md`, Section 11. The GBDT features are the gap, the card's history, the FSRS-6 default state replayed over that history, running user statistics, and a smoothed lexeme recall rate (Duolingo only, fitted on training rows). The model uses the same inputs as the other models, plus the lexeme id. It does not use `history_seen` or `history_correct`.

Selection used validation only: the last 20% of each user's training days, on 100 Anki users and a 5,000-user Duolingo development sample. The test set was used once for v3 and GBDT together, and the GBDT choice was fixed before that run. The v2 selection rule was revised afterwards on validation confidence intervals. Table 7 gives the test results on both datasets. Each difference is against FSRS-6 (optimised), with a 95% user-clustered paired bootstrap. The win column is the number of Anki users (out of 100) or the share of Duolingo users (out of 41,223) whose log loss is lower than FSRS-6 (optimised).

Table 7: Improved models on the test events of both datasets. The best log loss on each dataset is in bold. Differences are model minus FSRS-6 (optimised), with 95% user-clustered CIs.

	Anki (100 users)			Duolingo (all users)		
Model	Log loss	Diff [95% CI]	Wins	Log loss	Diff [95% CI]	Wins
FSRS-6 (default)	0.396	+0.017 [+0.008, +0.025]	28	0.302	+0.002 [+0.001, +0.002]	0.000
FSRS-6 (optimised)	0.379	reference	–	0.300	reference	–
HLR	0.680	+0.301 [+0.246, +0.361]	7	0.418	+0.118 [+0.112, +0.124]	0.577
Ours v1 (MCMC)	0.422	+0.043 [+0.033, +0.055]	22	0.268	−0.032 [−0.033, −0.031]	0.740
LagLogit + dt	not run	–	–	0.291	−0.009 [−0.012, −0.007]	0.716
Ours v2 (MAP)	0.379	−0.000 [−0.003, +0.003]	56	0.289	−0.011 [−0.012, −0.010]	0.704
Ours v3 (MAP)	0.378	−0.001 [−0.004, +0.002]	58	0.283	−0.017 [−0.018, −0.016]	0.715
GBDT (LightGBM)	**0.358**	**−0.021 [−0.028, −0.014]**	**82**	**0.259**	**−0.041 [−0.042, −0.041]**	**0.780**

GBDT is best on both datasets and on every metric (log loss, Brier, AUC, ECE, RMSE and mean bias), and its gap to FSRS-6 (optimised) is significant on both. The Bayesian models are competitive with FSRS-6 (optimised) but are not the best predictors. Much of the GBDT gain comes from the running user statistics and from training on all non-first reviews, not from a memory model. Stacking GBDT with v3 and FSRS-6 found no gain, so GBDT is the final model.

Disclosures for this section:

- **The test set was looked at twice.** v2 was tested in Phase 6. v3 and GBDT were tested together in Phase 7, after the GBDT choice was frozen on validation.
- **The v2 rule was revised post hoc.** The first rule chose v5, which wins only on Duolingo. The revised rule (win on both datasets, then lowest mean validation log loss) chose v2. Both changes used validation confidence intervals only.
- **Variants tried on validation.** v2: 9. v3: 23. GBDT: 38 in the first sweep (invalid, see below), then 16 on the fixed code. Stacking: 6. All are listed in the experiment logs.
- **A same-session leak was found in the first GBDT and fixed.** Duolingo logs a whole practice session at one timestamp, so the first running-statistics code let a row see simultaneous outcomes. Its validation log loss was too optimistic (0.2473 against 0.2578 after the fix, on the dev sample). Same-card duplicate rows at one timestamp are handled the same way. The invalid results are kept in `results/phase7_validation/invalid_leak/`.
- **The lexeme rate uses other users' training rows.** These rows can be later in calendar

time than a scored row. This is consistent with the per-user split and changes log loss by about 0.001 on Duolingo.

- **v2 and v3 are MAP fits**, not full posterior samples. Their intervals are the bootstrap intervals above, not posterior intervals.
- **The v1 MCMC model is still strong on Duolingo (0.268),** but its lambda does not converge there (Section on Duolingo above), so its lambda is not interpreted.
- **FSRS-6 (optimised) falls back to defaults for most Duolingo users** (115,110 of 115,222), so it is nearly the default model there.

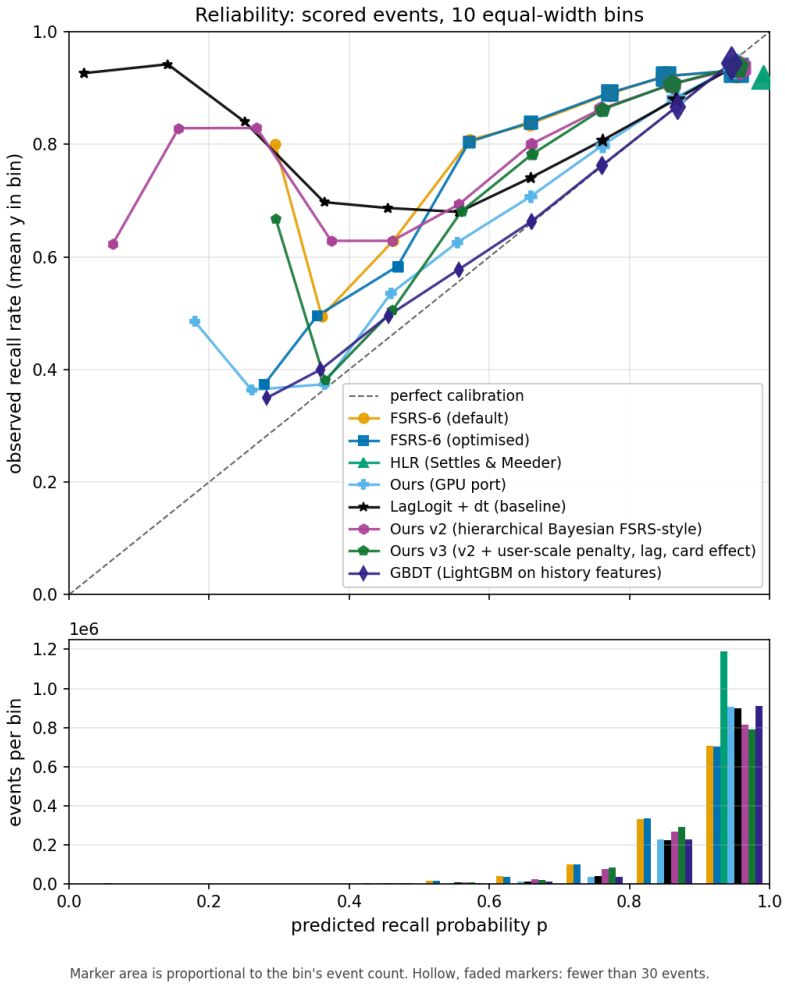


Figure 4: Duolingo (test, GBDT and all other models): reliability diagram over all gaps, from `results/duolingo_final`. Marker area is proportional to the bin count.

9 Discussion

- **FSRS is the stronger model on this data.** It is better in every gap bin, on discrimination and on calibration. FSRS also models a latent memory state per card (stability and difficulty), so it is not one of the static baselines our earlier updates compared against.
- **Comparison with earlier claims.** Against static baselines, the continuous-time state-space approach clearly helps: our model beats HLR by 0.258 in log loss. Against FSRS, however, our model shows no long-gap advantage. At 15–60 d and 61 d+, FSRS-6 (optimised) has lower log loss, higher AUC and better calibration. Any claim in the group paper that the DLM

fixes long-gap miscalibration should be limited to static baselines, or should report this result alongside it.

- **Where our model falls short.** It discriminates worse than FSRS (AUC 0.692 against 0.777), and it is over-confident at long gaps. Likely contributors are the non-converged decay rate, a single global λ (FSRS fits its parameters per user), and the fact that our model treats a "Hard" answer the same as "Good" or "Easy", whereas FSRS uses all four grades.
- κ**.** Practice-dependent decay slowing adds nothing measurable here.

10 Limitations

- **Convergence:** single-chain MCMC that did not converge (above). A run of 8,000–10,000 iterations with several chains and between-chain $\hat{R}$ is needed, at about 35 hours per 8,000 iterations on one CPU core.
- **Sample:** 100 users and at most 1,000 cards per user, one seed, one user sample. The bootstrap intervals reflect user sampling only, not run-to-run variation.
- **FSRS pipeline:** FSRS is run through `py-fsrs`, not the srs-benchmark pipeline (different split, no outlier filtering), so the numbers are not directly comparable with published leaderboards. The `py-fsrs` optimiser trains on at most the first 64 reviews of each card.
- **HLR:** no lexeme features, so this HLR is weaker than the original.
- **Scope:** same-day and first reviews are not scored. Cards with truncated histories are excluded.

11 Reproducibility

Code is in the `fsrs_benchmark/` package (commit `2eaa33c`, branch `BayesianScheduler`). It has 260 tests, which run with `pytest fsrs_benchmark/tests`. With a Hugging Face token that has accepted the dataset licence:

```
python -m fsrs_benchmark.run --run_name full_u100_c1000 --n_users 100 \
    --max_cards 1000 --seed 0 --iters 2000 --burn 1000 --workers 8
python -m fsrs_benchmark.report --run_dir results/full_u100_c1000
```

FSRS and HLR take about 15 minutes. Each variant of our model takes about 7.5 hours on one core. Every run records its arguments, package versions, git commit, counts and timings in `run_info.json`. A config fingerprint stops stale predictions from being reused. The full protocol and implementation notes are in `docs/EVALUATION.md`.

References

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